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Smart Energy Management and Demand Response

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Abstract

Demand Response (DR) programs, which incentivize electricity consumers to alter their usage patterns in response to grid conditions, represent a critical tool for enhancing grid stability and integrating renewable energy. The hospitality sector, characterized by significant and often flexible energy loads, is a prime candidate for DR participation. This review provides a comprehensive analysis of the state-of-the-art in demand response applications within hotels and resorts. It examines the key technologies and systems leveraged for load curtailment, with a detailed focus on high-impact areas such as Heating, Ventilation, and Air-Conditioning (HVAC), swimming pool pumps, lighting systems, and major appliances. The paper categorizes and discusses various DR strategies, from price-based load shifting to incentive-based, event-driven automated responses facilitated by Building Energy Management Systems (BEMS). Through an examination of international case studies, the review quantifies achievable load reductions, which often range from 10-30% of peak demand without compromising guest comfort. Furthermore, it addresses the primary challenges, including maintaining service quality, technology integration costs, and measurement and verification. Finally, the paper considers the regulatory landscape, including the role of building codes and specific initiatives in regions like Cyprus, concluding with an outlook on future trends such as deeper integration with renewables, AI-driven controls, and the aggregation of hotel loads into "virtual power plants."

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1 Introduction

Demand Response (DR) is defined as the modification of electricity consumption by end-use customers from their normal patterns in response to price signals or incentive payments, particularly during periods of high wholesale electricity prices or when system reliability is jeopardized [1]. Instead of activating additional peaking power plants, DR programs leverage large energy consumers as flexible, virtual resources to balance the grid [2, 3]. The hospitality sector, encompassing hotels, resorts, and other lodging facilities, presents a substantial and largely untapped opportunity for DR implementation. These facilities are characterized by significant energy consumption, predictable usage patterns tied to occupancy cycles, and centralized control over major energy-consuming systems, such as HVAC and water heating [2, 3].

The primary drivers for DR adoption in hospitality are twofold: rising energy costs and increasing pressure to adopt sustainable business practices [2, 3]. However, the paramount constraint is the non-negotiable requirement to maintain guest comfort and service quality; any DR measure must be effectively "invisible" to the guest experience [2, 3]. Early research and pilot programs established that, with meticulous planning, hotels can curtail 20-30% or more of their peak electric load for short durations without adversely affecting guests [2, 3, 4].

Contemporary DR applications in hospitality have advanced significantly, incorporating sophisticated automation, deep integration with Building Management Systems (BMS), and participation in a diverse range of market programs. These include both price-based DR, such as responding to time-of-use (TOU) tariffs, and incentive-based DR, which involves participation in utility-led load curtailment events. This review synthesizes the current body of knowledge on DR technologies, strategies, and programs implemented in the hospitality sector. It provides a detailed analysis of the primary systems targeted for load management, evaluates various DR program models, and highlights key findings from real-world case studies. The review also examines pertinent regulatory initiatives and discusses specific considerations for islanded grid systems, such as that of Cyprus.

This review draws on peer-reviewed literature, industry technical reports, and regulatory documents identified through targeted searches of IEEE Xplore, Scopus, and Google Scholar using terms related to demand response, hotel energy management, building automation, and HVAC optimisation. Additional sources include utility programme documentation, national energy plans, and technical standards. The review prioritises studies with empirical results from commercial building deployments, with particular attention to Mediterranean and island grid contexts relevant to the Cypriot operating environment.

2 Key Demand Response Technologies and Systems in Hotels

Hotels are complex ecosystems of energy-consuming systems, nearly all of which can contribute to demand response with appropriate controls and operational strategies. The most significant opportunities for load reduction are found in HVAC systems, water pumping, and lighting. Table 1 provides a summary of common DR measures across various hotel systems.

Hotel System	DR Measures	Comments/Impact (Illustrative)
HVAC - Chillers & A/C	• Raise zone temperature setpoints by 2-4°C. • Pre-cool spaces before peak hours. • Temporarily cycle off or reduce chiller compressor load. • Utilize thermal energy storage or building thermal mass. • Limit or cycle air handling unit fans via VFDs.	Largest load, often ~50% of hotel energy [5]. Central chillers can be curtailed for up to an hour with minimal comfort loss, shedding hundreds of kW [3, 4, 5]. Fan speed reduction offers non-linear power savings (e.g., 10% speed reduction ≈ 27% power saving) [6, 7].
Swimming Pool Pumps	• Shift filtration cycles to off-peak times. • Install and control variable-speed drives (VFDs). • Deactivate fountains and water features during events.	Pool pumps are significant, deferrable loads. OpenADR-enabled controllers can automatically reduce pump speed, cutting power by 50-60% during DR events with minimal impact on water quality [8, 9].
Lighting Systems	• Dim LED lighting by 20-40% in common areas. • Deactivate non-essential lighting (e.g., façade, signage). • Implement automated, centralized dimming tied to DR signals.	Lighting can account for 30-40% of commercial electricity use [1]. Dimming up to 30% can go unnoticed [1]. Automated systems can reduce lighting power by at least 15% across zones, as mandated by some codes [10].
Appliances & Equipment	• Reschedule laundry and dishwashing operations. • Cycle off spa heaters or non-critical pumps. • Park select elevators or reduce operating banks. • Deactivate decorative amenities (saunas, fountains).	Many operational processes are flexible. Hotels can shift laundry to off-peak hours [2, 3]. Case studies show resorts turning off fountains and reducing elevator service for 2-3 hours to meet DR targets [6, 7].
Building Control (BMS)	• Automatically execute pre-programmed load-shed plans. • Receive and interpret external DR signals (e.g., OpenADR). • Orchestrate complex sequences across multiple systems.	The BMS is the core enabler of advanced DR. It allows for rapid, reliable, and automated execution of curtailment strategies, freeing staff from manual intervention and enabling participation in fast-response programs [11].

Table 1 - Summary of Common DR Measures Across Various Hotel Systems

2.1 HVAC Systems and Chillers

HVAC systems are the single largest DR resource in most hotels, responsible for over 50% of a building's total energy consumption, with cooling loads driving summer grid peaks [5]. The centralized nature of hotel chiller plants and air handling units makes them highly amenable to aggregate control. Key DR strategies for HVAC include:

- **Global Setpoint Adjustment:** Raising cooling setpoints by 2-4°C across common areas and unoccupied rooms is a simple yet effective measure. Modern demand-responsive thermostats can automate this process upon receiving a DR signal [10].
- **Pre-cooling and Thermal Mass:** Leveraging the building's thermal inertia is a cornerstone of HVAC DR. By pre-cooling spaces during off-peak hours, the HVAC system can be significantly reduced or turned off during a peak event, allowing the building's temperature to "coast" upwards slowly without impacting comfort [5].
- **Direct Chiller Control:** For short durations (30-60 minutes), chillers or compressors can be cycled or shut down entirely, yielding substantial and rapid demand reductions. Field experiments have demonstrated load curtailment of 380-459 kW within 15 minutes via this method, with minimal degradation of occupant comfort [5].
- **Ancillary Services:** Advanced control strategies enable hotel HVAC systems to provide grid services beyond simple load shedding. Research has demonstrated the feasibility of using aggregated air conditioning units for fast-frequency regulation, where compressors are shut off for seconds at a time to arrest grid frequency deviations [4] [12].

Regulatory mandates are accelerating the adoption of DR-ready HVAC controls. California's Title 24 energy code, for instance, requires that new large-scale HVAC systems be capable of remotely adjusting cooling setpoints by at least 4°F (~2.2°C) during a DR event [10].

2.2 Swimming Pool Pumps and Water Systems

Swimming pool pumps, fountains, and other water features represent a significant source of flexible, deferrable load. The timing of water filtration is often not critical, allowing pump operations to be shifted to off-peak hours or temporarily curtailed. A California pilot project successfully demonstrated the use of OpenADR-compliant controllers on variable-speed pool pumps. During DR events, these controllers automatically triggered a pre-programmed low-speed mode, significantly reducing power consumption [8]. The study concluded that shifting pool pump operation could meaningfully reduce afternoon peak demand [8, 9]. Similarly, decorative fountains and spa heaters can be deactivated during DR events with little to no guest impact, and were included as recommended actions in utility guides for hotels [13].

2.3 Lighting and Electrical Systems

Lighting constitutes another major electrical load, and its controllability makes it an excellent DR resource. Active DR for lighting involves deliberately reducing illumination in response to a grid signal. Studies indicate that dimming levels of 15-30% are often imperceptible to occupants, especially when implemented gradually or in daylit areas [1]. Automated systems can execute these changes seamlessly. Building codes are again a driving factor; California's Title 24 requires

that new lighting systems exceeding 4 kW have controls capable of automatically reducing power by at least 15% [10]. In practice, hotels typically focus on dimming in common areas and deactivating non-essential exterior and decorative lighting, while ensuring safety-critical lighting remains unaffected [2, 3] [10].

2.4 Appliances and Other Flexible Loads

A variety of other operational loads offer flexibility for DR.

- **On-premise laundries**, with their high-power washers and dryers, are prime candidates for load shifting. Hotels can reschedule laundry operations to evening or overnight hours to avoid peak periods [2].
- **Kitchen equipment**, such as dishwashers, can have their cycles postponed for short durations.
- **Elevators** in high-rise hotels can be managed by taking one or more cars out of service temporarily, a strategy employed by a large resort to achieve its DR target [6, 7].

The common thread for these loads is rescheduling or selective shutdown of non-guest-facing or non-essential equipment. While often requiring manual coordination, their collective contribution to a hotel's overall load reduction can be significant.

3 Automation, Energy Management, and DR Programme Structures

The orchestration of diverse DR measures across a large facility necessitates a centralized control platform. The Building Energy Management System (BEMS or BMS) is the critical enabler for sophisticated, reliable DR. Automated Demand Response (ADR) leverages the BEMS to execute pre-programmed load curtailment strategies automatically upon receiving a signal from a utility or DR aggregator [11].

The OpenADR protocol has emerged as an industry standard for communicating DR event information. A BEMS equipped with an OpenADR client can receive signals and trigger a sequence of actions-such as adjusting HVAC setpoints, dimming lights, and cycling pumps-within seconds and without human intervention [10] [14]. The benefits of ADR are substantial. First, automation ensures a rapid, consistent, and verifiable response, which is essential for participation in fast-response ancillary service markets [11]. Second, it eliminates the need for facility staff to perform manual adjustments during an event, a process that is slow and prone to error [11]. Third, the BEMS logs performance data, which is crucial for verifying load reduction, validating programme participation, and optimizing future DR strategies. The industry trend is overwhelmingly toward automation, with DR providers like Enel X noting that it allows companies to “not even have to lift a finger” to participate [11].

The degree of automation is closely tied to the type of DR programme a hotel participates in. Hotels engage in demand response through two primary channels: price-based programmes and incentive-based programmes [1, 3].

Price-based DR is a voluntary response to dynamic electricity pricing. Hotels on Time-of-Use (TOU) or Critical Peak Pricing (CPP) tariffs strategically shift load to minimize costs. This is fundamentally an economic optimization practice. Common strategies include pre-cooling the building before the high-price afternoon peak and shifting deferrable loads like laundry and pool filtration to cheaper, off-peak hours [5]. This form of DR is an extension of daily energy management and provides cost savings independent of any formal incentive payments. In markets with real-time pricing, the potential savings increase further, as the BEMS can continuously optimise consumption against fluctuating wholesale rates rather than responding to fixed time-of-use bands [3].

Incentive-based DR, by contrast, involves a contractual agreement where a hotel is compensated for curtailing load upon request from a utility or aggregator. These programmes are designed to ensure grid reliability during periods of stress. Key program types include:

- **Capacity Programs:** Hotels commit to a specific kW load reduction capability and receive regular capacity payments (e.g., $/kW-year) for being available, plus potential energy payments for actual performance during events [6].
- **Demand Bidding Programs:** Hotels can bid their load reduction capability into energy markets, receiving the market price for energy when their bid is accepted [4].
- **Emergency/Reliability Programs:** These require very fast response to critical grid events. While historically dominated by industrial loads, aggregated hotel HVAC systems are increasingly seen as a viable resource for these high-value services [15] [16, 14].

Participation is often facilitated by a third-party aggregator or Curtailment Service Provider (CSP), who manages the programme enrollment, signaling, and verification on behalf of the hotel [2]. While manual response is possible, participation in most modern incentive programmes, particularly those with short dispatch times, effectively requires the automation infrastructure described above [11]. The interplay between programme type and automation level is significant: price-based DR can be partially implemented with manual schedule adjustments, but realising the full economic potential of dynamic pricing requires automated, real-time control. Incentive-based programmes, especially those providing ancillary services, are effectively inaccessible without a fully automated BEMS [9] [14].

4 Implementation Case Studies

Real-world implementations validate the feasibility and benefits of DR in the hospitality sector. The following cases, drawn from a range of geographies and hotel types, illustrate the diversity of approaches and the consistency of results.

- **Seaport Boston Hotel (USA):** An early adopter, this luxury hotel enrolled in a DR programme in 2009. By adjusting HVAC, reducing lighting, and pausing laundry operations via its BMS, the hotel consistently achieves a load reduction of over 300 kW during events. The programme generates thousands of dollars in annual revenue, which is reinvested into sustainability initiatives, all while remaining "invisible" to guests [2, 3].
- **Large Hotel & Water Park Resort (Ohio, USA):** This facility developed a 400 kW curtailment plan involving chiller adjustments, idling elevators, and slowing VFD-controlled fans. The hotel earned $14,000-$17,000 annually in capacity payments alone for this capability [6, 7]. This case highlights the strategy of combining multiple smaller reductions to achieve a large aggregate target.
- **Marriott Select-Service Hotels (California, USA):** A portfolio of 14 smaller Marriott hotels installed EMS with Auto-DR capabilities, supported by utility incentives. Participating through an aggregator, their primary DR action is automatically adjusting public space HVAC setpoints after pre-cooling. This case demonstrates the scalability of DR across a hotel chain and the importance of utility-sponsored enabling technology programmes [17].
- **UK Hotels in National Grid STOR Programme:** Research involving large UK hotels demonstrated their ability to participate in the Short-Term Operating Reserve (STOR), a critical grid ancillary service. By equipping chillers with high-resolution metering and controls, the hotels could provide rapid, verified load drops within minutes of a grid request, proving that hospitality assets can function as high-value grid resources [16, 14].
- **Mediterranean and Island Grid Contexts:** While the case studies above are drawn from mature DR markets, research on island and Mediterranean hotel systems confirms the applicability of these approaches to smaller, isolated grids. Meschede [18] analysed the DR potential of hotels in the Canary Islands, demonstrating that probabilistic scheduling of hotel loads - accounting for the stochastic nature of occupancy, weather, and tourism seasonality - can yield substantial peak reduction potential even in the absence of formal DR market structures. Similarly, Beccali et al. [19] characterised end-user electricity consumption on the Mediterranean island of Lampedusa, identifying the seasonal concentration of hotel loads as both a grid challenge and a DR opportunity. These findings are directly relevant to the Cypriot context, where hotel demand follows comparable seasonal patterns and the isolated grid faces similar integration challenges.

These cases collectively demonstrate that hotels can reliably shed 10-30% of their peak load, with financial and sustainability benefits, by leveraging a combination of automated controls and well-defined operational plans. The evidence base spans from large urban luxury properties to portfolios of smaller select-service hotels, and from mature continental DR markets to island grid systems - confirming that the core principles are transferable across contexts and scales.

5 Challenges, Regulatory Context, and the Case of Cyprus

Despite its demonstrated benefits, DR implementation in hotels faces several practical challenges that must be addressed for successful deployment.

Guest comfort remains the paramount constraint. All DR strategies must be designed and executed to avoid any negative impact on the guest experience [2, 4]. This requires careful calibration of setpoint adjustments, conservative load-shed targets, and robust monitoring to detect and respond to comfort threshold breaches before they affect guests. The case studies reviewed in Section 5 consistently emphasise that successful hotel DR programmes are those where the load reduction is "invisible" to guests - a standard that demands both sophisticated control logic and real-time feedback loops.

Technology and cost present a second barrier. Retrofitting older properties with the necessary BMS, smart controls, and sub-metering can be cost-prohibitive without incentives, particularly for small and medium-sized hotels that lack the capital budgets of large chains [1] [17] . The cost barrier is compounded by the need for specialised integration work to connect legacy HVAC systems with modern automation platforms. However, the case studies also demonstrate that utility-sponsored incentive programmes and third-party aggregator models can substantially reduce the financial burden on individual hotels.

Operational integration poses additional difficulties. Aligning DR events with hotel operations requires flexibility: a high-occupancy period or a major hotel function may require the hotel to opt-out of an event [16, 14]. This is addressed in well-designed programmes through opt-out provisions that allow hotels to decline events a limited number of times per season without losing their capacity payments. Measurement and verification (M&V) is another area of complexity: accurately establishing a baseline of what energy use would have been absent the DR intervention requires statistical modelling and careful data management, and efficient buildings may find it harder to demonstrate large savings relative to their already-low baseline. Finally, proper staff training is required to ensure that hotel personnel understand the DR protocols and do not inadvertently override automated actions during events.

Regulatory frameworks are a key driver of DR adoption and are increasingly mandating DR-ready technologies. Building energy codes, such as California's Title 24, require new construction to include DR-capable HVAC and lighting controls [10]. Green building certifications like LEED offer credits for DR capability, further incentivizing designers and building owners [1]. At the European level, the Clean Energy for All Europeans package and the recast Electricity Directive (EU 2019/944) establish the right of consumers, including commercial consumers, to participate in DR and flexibility markets, and require member states to remove regulatory barriers to demand-side participation [3].

For an islanded grid system like Cyprus, DR holds immense strategic value. Isolated grids are more vulnerable to frequency disturbances and often rely on expensive imported fossil fuels for generation, making peak demand management both a reliability and an economic priority. The Cyprus Integrated National Energy and Climate Plan (INECP) recognizes this, estimating an "untapped demand response potential of around 50 MW by 2030" [20]. Achieving this will be

critical for managing summer peak loads, driven largely by air conditioning in the tourism sector, and for integrating the country's growing solar PV capacity.

Currently, Cyprus is in the early stages of developing a formal DR market, with plans for a wide-scale smart meter rollout and regulatory updates to comply with EU directives on grid flexibility [20] [21]. The Cyprus Energy Regulatory Authority (CERA) has initiated market reform processes aimed at introducing competitive electricity supply and enabling demand-side participation, though the timeline for a fully operational flexibility market remains uncertain. Historically, Cyprus has utilized demand-side resources in emergencies, such as enlisting customer-owned generators into a "virtual power plant" following the Vasilikos power station explosion in 2011 [22]. This precedent demonstrates a foundational capability for demand-side participation. As the formal market develops, hotels and resorts, as some of the island's largest commercial electricity consumers, will be natural and essential participants in future DR programmes [19] [18].

Beyond the academic literature and utility-sponsored pilot programmes reviewed above, the commercial frontier has moved beyond one-off demand response pilots into software-led continuous optimisation. A useful distinction is between control-layer companies, which optimise building systems in real time, and market-layer aggregators, which monetise flexibility in grid and capacity markets.

BrainBox AI[1] is one of the clearest examples of the first category: its platform applies predictive AI to HVAC operation in existing commercial buildings, with claimed reductions of up to 25% in energy use and up to 40% in associated greenhouse-gas emissions [23]. The strategic significance of this model is underscored by its acquisition by Trane Technologies in January 2025, suggesting that autonomous HVAC control is now being absorbed into mainstream building-services infrastructure rather than remaining a niche offering. In parallel, firms such as Parity[2] are demonstrating a more operations-oriented route to market, particularly in multifamily and hospitality environments, where the value proposition combines day-to-day HVAC optimisation, automated demand response participation, and compliance with tightening building-performance standards [24]. Parity positions this not as a capital-intensive retrofit, but as a software-and-service layer over existing HVAC assets, reporting both utility-bill savings and demand response revenues from deployed properties.

The second frontier is the integration of flexible building loads into wider electricity-market participation. In the UK, Flexitricity[3] illustrates how commercial flexibility has evolved from manual curtailment into an algorithmically traded grid resource: the company operates as Britain's first open-market demand response aggregator, reporting 540 MW of flexible capacity under management and participating in services such as STOR, the Capacity Market, and the Balancing Mechanism [25]. This matters for hotels because it demonstrates that flexibility is increasingly monetised not only through avoided consumption, but through stacked revenue streams tied to system balancing and reserve provision.

[1] https://brainboxai.com
[2] https://www.paritygo.com
[3] https://www.flexitricity.com

The implication for hospitality is that the cutting edge is no longer simply smart control within the building boundary, but the coupling of building automation, forecasting, and market access into a single commercial stack. The most advanced implementations increasingly treat HVAC plants, pumps, storage, and other controllable hotel loads as dispatchable economic assets, not just efficiency opportunities. Cyprus currently lacks any comparable commercial offering — the gap between this international commercial frontier and the Cypriot status quo is the subject of the following section.

The confluence of regulatory pressure (from the EU framework), economic incentive (from rising energy costs and future flexibility market revenues), and technical opportunity (from the maturity of BEMS and DR technologies documented in this review) creates a favourable environment for DR deployment in Cypriot hotels. What is currently missing is the integration layer - the AI-driven energy management and automated control systems that would enable these facilities to participate in DR programmes as they emerge. This gap is the focus of the following section.

6 Energy Management in Cypriot Hotels: Current State and Gap Analysis

The preceding sections of this review have established that international best practice in hotel energy management centres on automated DR systems, AI-driven optimisation, and deep integration between building management platforms and grid signals. To assess how the Cypriot hospitality sector compares against this benchmark, the project team conducted a structured assessment drawing on published data, industry contacts, and direct observations from the pilot site engagement with Elias Beach Hotel. The findings confirm a substantial gap between current local practice and the capabilities documented in the international literature.

The majority of Cypriot hotels, including large coastal properties that are among the island's most energy-intensive commercial consumers, continue to rely on basic building management configurations. Typical installations are limited to keycard-based room occupancy controls and fixed-schedule HVAC programming that does not respond to real-time conditions such as actual occupancy levels, outdoor temperature fluctuations, or electricity price signals. Where centralised BMS platforms exist, they are generally used for monitoring rather than active control, and none of the facilities assessed were found to have any form of automated demand response capability. This stands in contrast to the OpenADR-enabled, BEMS-orchestrated systems described in Sections 4 and 6 of this review.

The absence of DR infrastructure has several compounding effects. First, hotels cannot participate in grid-stabilising programmes or benefit from the capacity payments and cost savings documented in the case studies reviewed in Section 5. Second, the lack of real-time energy analytics means that significant optimisation opportunities - particularly in HVAC pre-cooling, pool pump scheduling, and lighting modulation - remain unexploited. Third, the growing installed base of solar photovoltaic systems on Cypriot hotel rooftops is not being managed dynamically; self-consumption is not optimised against time-varying load profiles, and excess generation is simply exported at net-metering rates rather than being used to offset peak consumption through intelligent scheduling. This is particularly significant given that Cyprus has experienced over 120 GWh of renewable energy curtailment at the distribution system operator level, representing substantial lost economic value [15].

Table 2 below summarises the gap assessment across the key capability dimensions identified in this review. The pattern is consistent: capabilities that are well-established in mature DR markets remain either absent or at an early stage in Cyprus. This gap represents a significant opportunity for the SEMDR project, which targets precisely these capability deficits through the deployment of AI-driven energy management and automated demand response at the pilot hotel site.

Capability	International Best Practice	Current State in Cyprus	Gap Severity
Automated DR Response	OpenADR-enabled BEMS with sub-30-second response times [8] [9]	Manual adjustments by facility staff; no automated response capability	High
Real-Time Energy Analytics	ML-driven prediction and multi-objective optimisation [4] [14]	Basic monitoring with fixed schedules; no predictive analytics	High
Renewable Energy Integration	Dynamic self-consumption optimisation aligned with load profiles	Fixed net-metering; no dynamic management of PV generation	Medium
Grid Service Participation	Aggregated hotel loads providing ancillary services and capacity [14]	No formal DR market; no hotel participation in grid programmes	High
Comfort-Aware Load Control	Setpoint adjustment within validated comfort bands (±1.5-2°C) [3] [8]	No comfort-aware automated control; manual overrides only	High

Table 2 - Gap Assessment Across the Key Capability Dimensions Summary

7 Algorithmic Approaches to DR Optimisation

Beyond the physical DR measures and programme structures reviewed above, there is a growing body of research on the algorithmic methods used to optimise building energy consumption in response to dynamic price signals and grid conditions. These approaches range from simple rule-based heuristics to sophisticated model-based and data-driven techniques, and the choice of method depends on the available data, the controllable load characteristics, and the required speed of decision-making.

The simplest approaches use predetermined rules and schedules to shift or curtail loads. For example, a pool pump controller might apply a greedy algorithm that evaluates electricity price slots and selectively deactivates the pump during the most expensive periods, subject to water quality and minimum runtime constraints. These methods are transparent, computationally lightweight, and well-suited to loads with simple operating constraints where the primary objective is cost minimisation under known tariff schedules. Their principal limitation is that they cannot anticipate future conditions or optimise across multiple interacting systems; they are reactive by nature.

More advanced approaches employ Model Predictive Control (MPC) or Dynamic Programming (DP) to optimise energy consumption over a rolling time horizon [24]. In the context of HVAC systems, these methods model the building's thermal dynamics and solve an optimisation problem at each control interval, determining setpoints that minimise energy cost while satisfying comfort constraints [27]. DP-based approaches discretise the state space (e.g., temperature bins and control actions) and evaluate all feasible trajectories to find the globally optimal schedule. These methods can incorporate multi-term cost functions that balance energy expenditure, comfort violations, equipment cycling penalties, and peak demand charges. The principal advantage of MPC and DP is their ability to look ahead and make proactive decisions - for instance, pre-cooling a building in anticipation of a price spike - rather than simply reacting to current conditions. However, they require an accurate model of the building's thermal behaviour, which can be difficult and expensive to develop for existing buildings where detailed construction and equipment data may not be available [24].

Machine learning (ML) techniques are increasingly used to address this modelling challenge by building predictive models of building energy behaviour directly from operational data [25]. Gradient boosting methods such as LightGBM and XGBoost have shown strong performance in predicting thermal transitions and energy consumption as functions of outdoor conditions, occupancy, and control inputs. These ML models can serve as fast surrogate models within optimisation frameworks, replacing physics-based simulations with data-driven approximations that enable real-time decision-making [26]. Quality gates - such as minimum R^2 thresholds and maximum RMSE tolerances - are applied to ensure model reliability before deployment. Gaussian Process models offer an alternative approach that provides uncertainty estimates alongside predictions, which can be valuable for risk-aware control strategies. Recent work has also demonstrated the potential of physics-constrained deep learning for modelling multi-zone building thermal dynamics, combining the flexibility of neural networks with physical consistency guarantees derived from energy balance equations [26].

The integration of ML-based prediction with DP or MPC optimisation represents the current frontier in building energy management [24] [27]. This hybrid approach combines the predictive accuracy of data-driven models with the global optimality guarantees of systematic optimisation methods, enabling automated systems that can proactively manage energy consumption in response to price signals, weather forecasts, and occupancy patterns while maintaining strict comfort and safety constraints.

Table 3 provides a comparative summary of the three algorithmic families discussed above, evaluated against criteria relevant to hotel DR applications. For HVAC systems with complex thermal dynamics and multi-hour planning horizons, MPC and DP approaches offer the strongest theoretical guarantees, but require accurate thermal models that may be difficult to obtain for existing buildings. ML-based surrogate models address this limitation by learning thermal behaviour directly from operational data, enabling the hybrid ML+DP approach that the SEMDR project adopts. For simpler loads such as pool pumps, where operating constraints are straightforward and the objective is primarily cost minimisation under known tariff schedules, rule-based methods remain effective and computationally efficient.

Method	Computational Cost	Data Requirements	Optimality	Hotel DR Suitability
Rule-Based / Heuristic	Low	Tariff schedules, basic operating constraints	Local (greedy)	Good for simple loads (pool pumps, lighting)
MPC / Dynamic Programming	Medium-High	Thermal model, forecasts, state discretisation	Global (within model accuracy)	Strong for HVAC with known building dynamics
ML + Optimisation (Hybrid)	Medium	Historical operational data, weather, occupancy	Near-optimal (data-dependent)	Preferred for existing buildings with limited physical models

Table 3 - Comparative Assessment of Algorithmic Approaches for Hotel DR Optimisation

8 Conclusion and Future Outlook

Demand response in the hospitality sector has matured from a novel concept into a proven, valuable strategy for energy management. The convergence of advanced automation, data analytics, and supportive regulatory policies has enabled hotels to become active, dynamic partners in ensuring grid reliability. The state-of-the-art involves a holistic approach, where a central BEMS orchestrates load reductions across HVAC, lighting, and other systems in a manner that is both effective and imperceptible to guests.

The future of hospitality DR is likely to be shaped by several key trends:

- **Deeper Integration with Renewables:** DR will be used to align hotel energy consumption with the variable output of on-site and grid-scale renewables.
- **AI-Driven Optimization:** Artificial intelligence will enable more precise, predictive, and adaptive control of building loads, maximizing DR potential without crossing comfort thresholds.
- **Aggregation and Virtual Power Plants:** The aggregation of hundreds of hotels into a single, dispatchable "virtual power plant" will become more common, allowing the hospitality sector to participate in wholesale energy and ancillary service markets.
- **Policy and Market Evolution:** More jurisdictions will follow the lead of pioneers like California in mandating DR-readiness, while electricity markets will continue to evolve to better compensate flexibility.

For regions like Cyprus, the development of a robust DR market is not just an opportunity but a necessity for achieving a secure, cost-effective, and sustainable energy future. Hotels, as cornerstone consumers, are poised to play a pivotal role in this transition. By embracing the technologies and strategies outlined in this review, the hospitality sector can continue to enhance its operational efficiency, generate new revenue streams, and contribute meaningfully to the stability and sustainability of the electric grid.

The findings of this review have directly informed the design of the SEMDR (Smart Energy Management and Demand Response) system, which aims to demonstrate these advanced technologies in the Cypriot hospitality context. By targeting HVAC and swimming pool pump optimisation as priority DR applications - the two largest flexible loads identified in this review - and by incorporating algorithmic approaches ranging from rule-based scheduling to ML-assisted dynamic programming, the SEMDR project addresses the specific gaps identified in the Cypriot energy management landscape while building on proven international best practices.

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